

4. (a) State the meaning of the term diffusion. [1]

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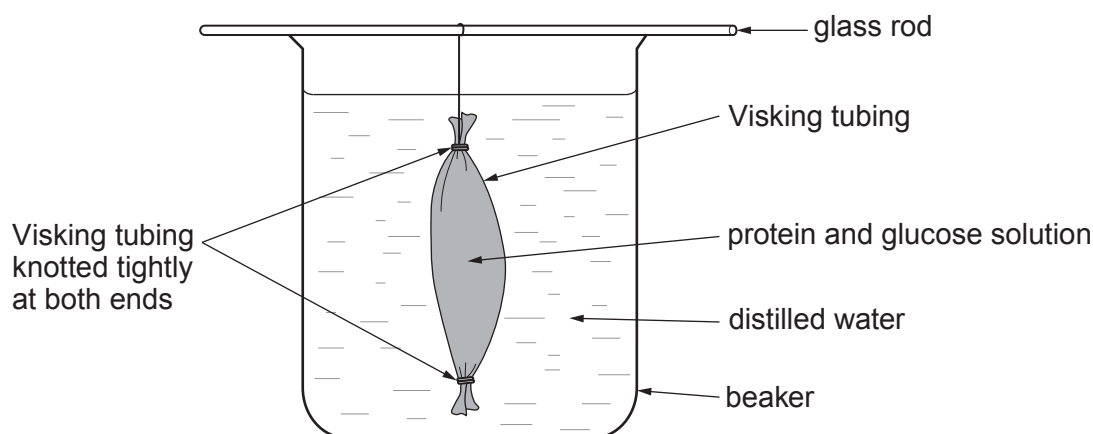
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(b) After a lesson on the properties of cell membranes a year 10 class was asked to investigate some of these properties using Visking tubing. They were given the following instructions:

- Soak a 15 cm length of Visking tubing in water to soften it.
- Tie a knot in one end of the tube.
- Fill the tube with a solution made up of protein and glucose dissolved in water.
- Tie a knot in the open end of the tube.
- Wash the tube under a stream of tap water for 15 seconds.
- Using a glass rod suspend the Visking tubing in a beaker of distilled water.



The diagram below shows how your apparatus should appear.



(i) Why were the students instructed to 'wash the tube under a stream of tap water for 15 seconds'? [1]

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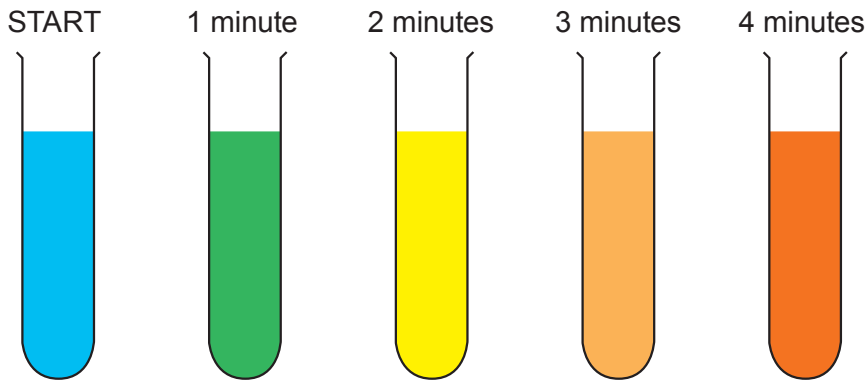
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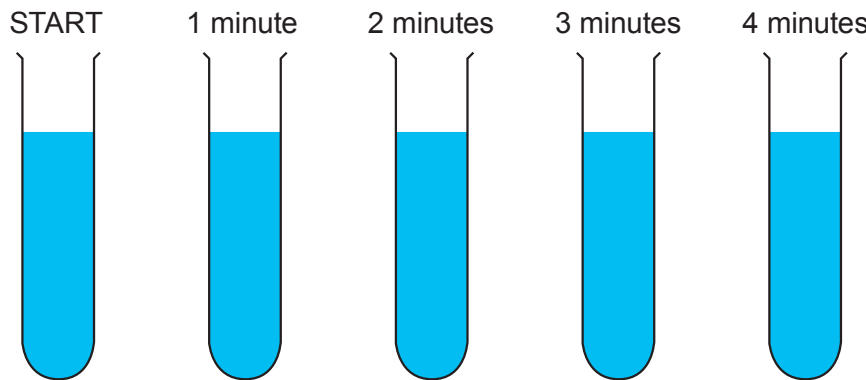
Examiner
only

The students were asked to sample the distilled water **in the beaker** for the presence of both glucose and protein at the start of the experiment and every minute for the next four minutes. Dafydd decided to photograph his results on his smart phone. The photographs he obtained are shown below.

Benedict’s test for glucose



Biuret test for protein



(ii) Explain Dafydd’s results, **at four minutes**, for the contents of the **beaker** for both the Benedict’s test and the Biuret test.

Benedict’s test [3]

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Biuret test [3]

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